Virtual Vials

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Project Overview

- Virtual Vials will primarily focus on developing a VR-based learning environment for Newman Projections. It will involve users simulating a laboratory setting to enable interactive visualization of molecular structures.

Problem and Research Goal

Project Goal

- Virtual Vials aims to offer an interactive and engaging VR educational tool for chemistry students
- Virtual vials should be a suitable addition to a first-to-second-year university chemistry student's learning, primarily for a molecule's Newman projection in a 3D space

Problem

- Learning Newman Projection may be hard to grasp initially
- Some students find it hard to visualize molecules

Motivation

- We want to offer an educational tool to fill the gap between textbook material and a physical lab with VR experience.

Objectives and Approach

Method Overview

- Builder - using a molecule bank, users can pick an atom and build any molecules to their liking.
- Newman Projection - as the player builds a molecule, the corresponding Newman Projection is displayed.

Tools and Libraries

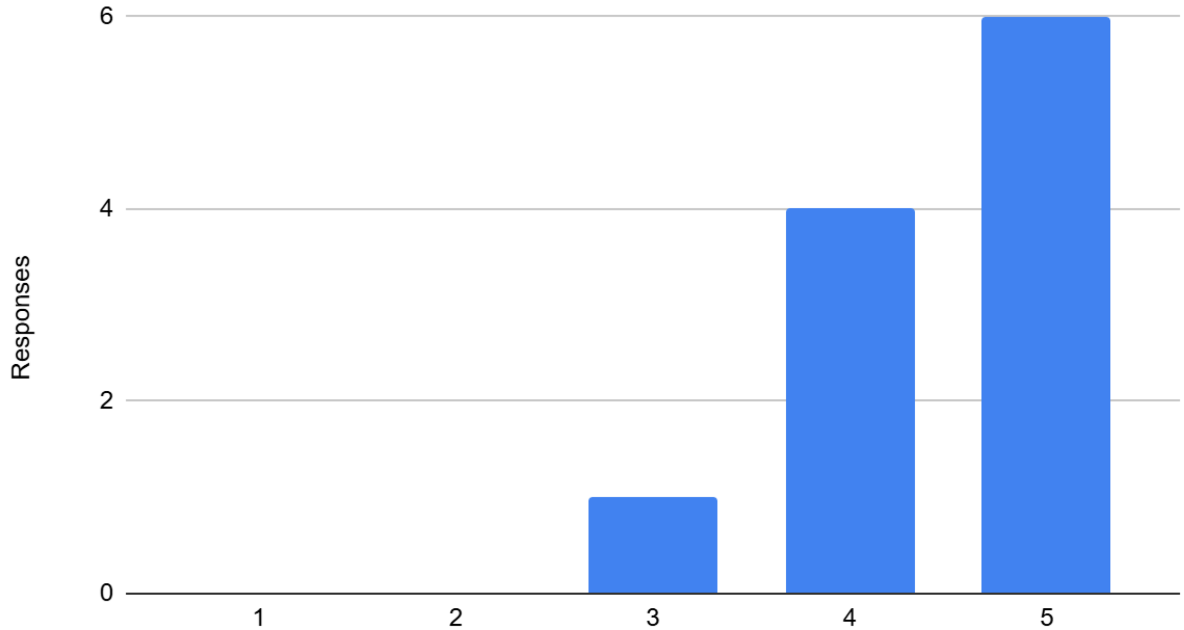
- Meta Quest 3s - VR device used to run the app
- Unity 6 - game engine used to develop the entire project
- XR Interaction Toolkit - Unity package that provided us with templates for all VR interactions
- Blender - used to make some of the 3D assets in the game
- Google Forms - used to make the surveys for user testing

Work Completed

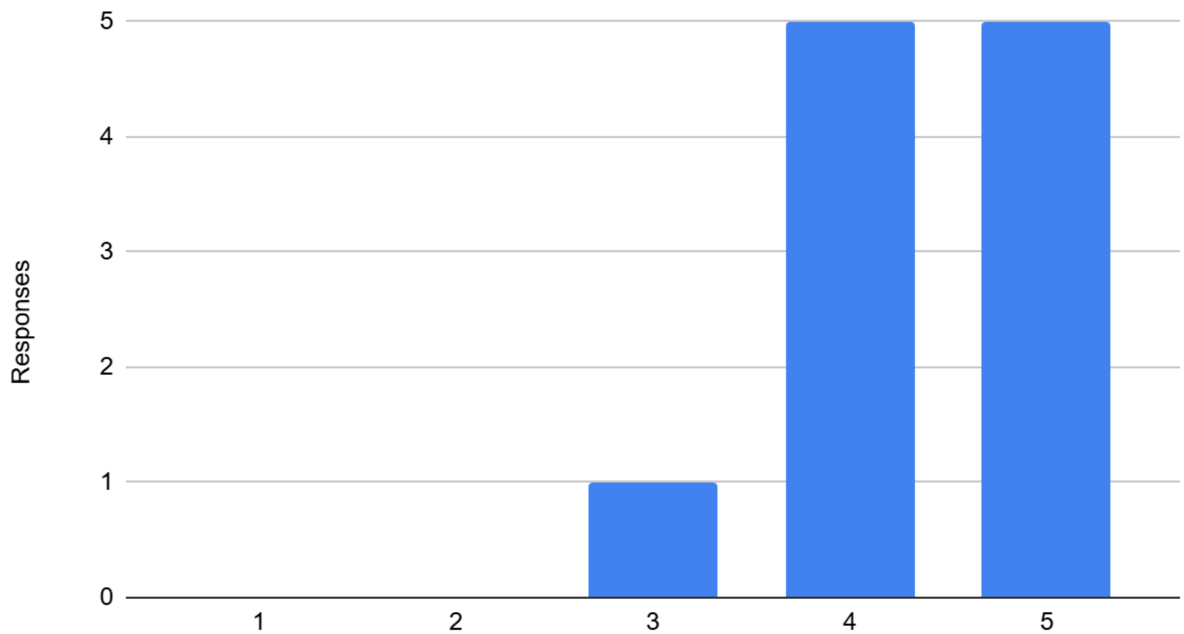
Data Collected

- To test the app's effectiveness and usability, we let 11 computer science students playtest, and the following are the results:

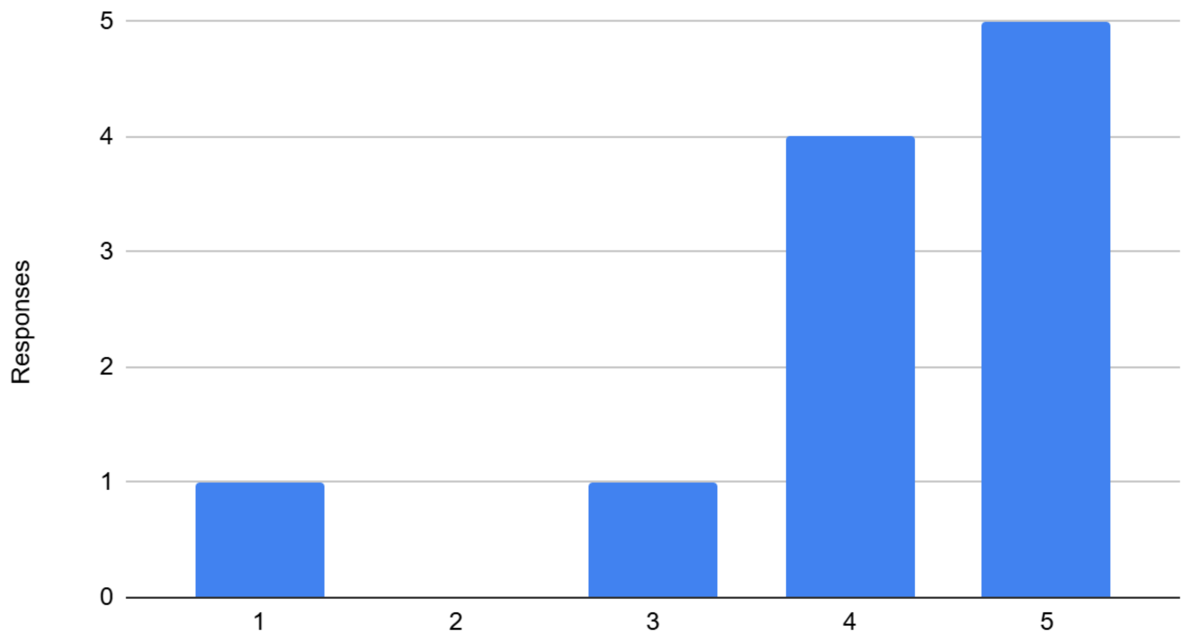
Instructions were clear and easy to follow



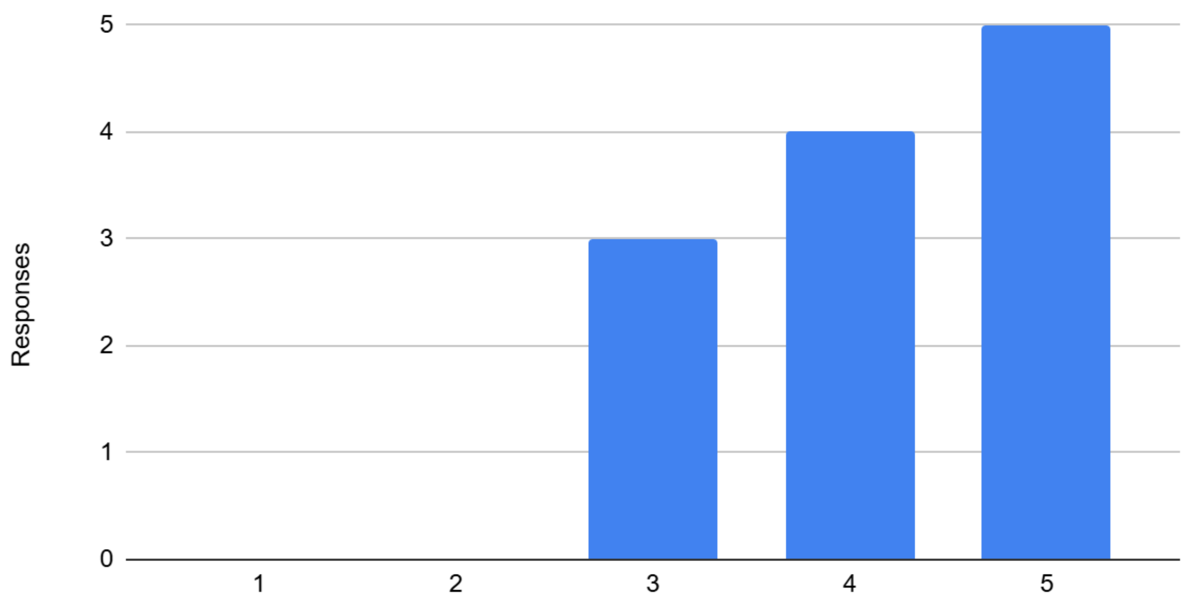
The controls were intuitive and easy to use.



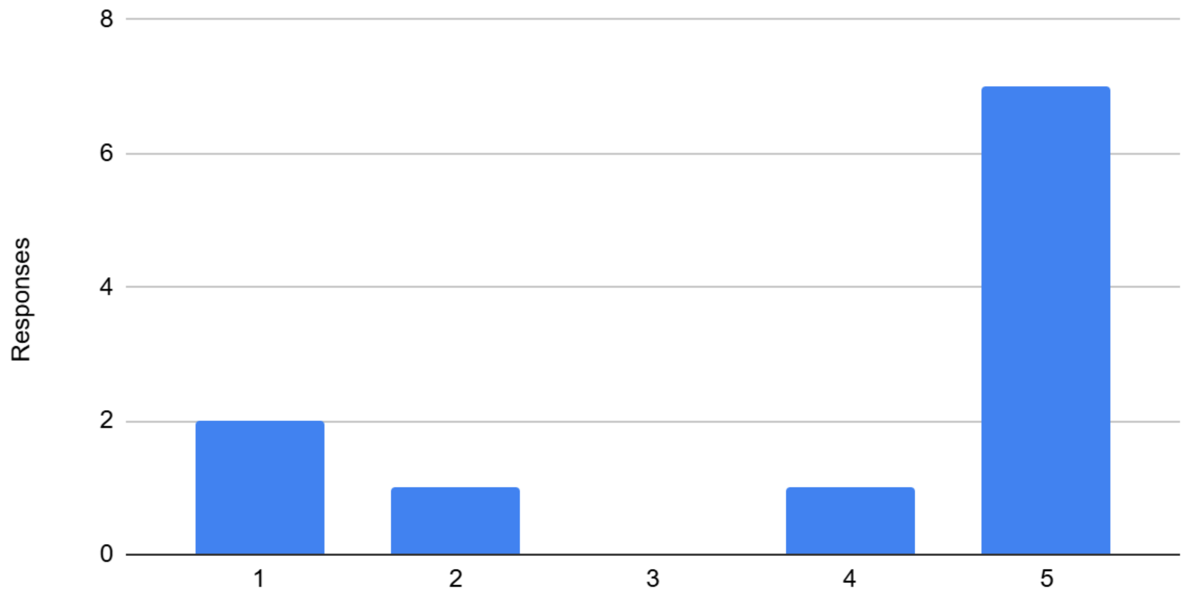
I would rather use a VR tool than the traditional 2D format.



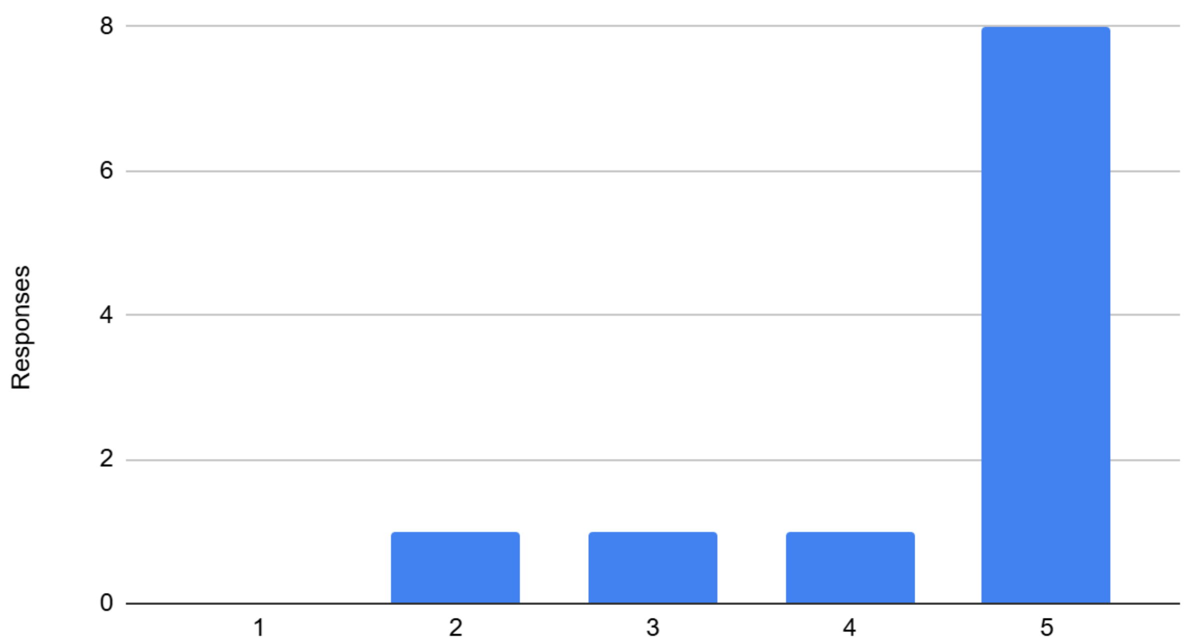
Learning with this tool or a similar one would help me learn concepts more quickly.



I did not experience motion sickness or discomfort while using the app.



Would you use this app again?



- According to the data presented, the app was a success, though some people experienced discomfort, which is to be expected given that it was their first time with VR.

Zhire’s Work

- My role in the project focused primarily on the game's visuals, which included all the UI elements, assets, and scenes. Additionally, I implemented gameplay features for all VR interactions and SFX to ensure good user flow. I also worked on one of the core features of the app, which was the spawning of the atoms. I worked with the team to ensure the spawned atoms were accurate and functioned as intended. I also worked on the early version of the Builder functionality, combining prefabs with XR socket and grab interactions.

James’ Work

- My role in Virtual Vials was primarily focused on everything related to the Newman projection of the molecule, including the projection of the ready-made molecule in the visual room, the initial rotation of the molecule around a bond in a previous implementation, and the projection of built molecules in the builder room. Additionally, I implemented the bond hovering and selection for displaying the projection of the bond. The projection included taking the bond and molecules and their attached molecules in their 3D space and projecting them onto a 2D plane according to their angles from the front/back carbon according to standard Newman Projections.

Dahlin’s Work

- My role in the project was focused on the molecule builder. This system included components to manage the bonding of several different types of atoms, components to manage picking up and moving atoms and molecules, as well as components to manage the geometry of combined molecules and atoms. Early on in the project I also had a couple minor contributions to implementations of the UI.

Demo

- https://youtu.be/VpAywnsbLWw

Github Link

- https://github.com/Dahlin-W/VirtualVials

Gameplay Mechanics

Mechanics

- Virtual Vials have two main components: Builder and Newman Projections.
- Builder - with the builder, the player can spawn atoms that they can use to build molecules.
- Newman Projections - as the player builds, the corresponding Newman projection of the molecule is dynamically shown in one of the UI panels.

Controls

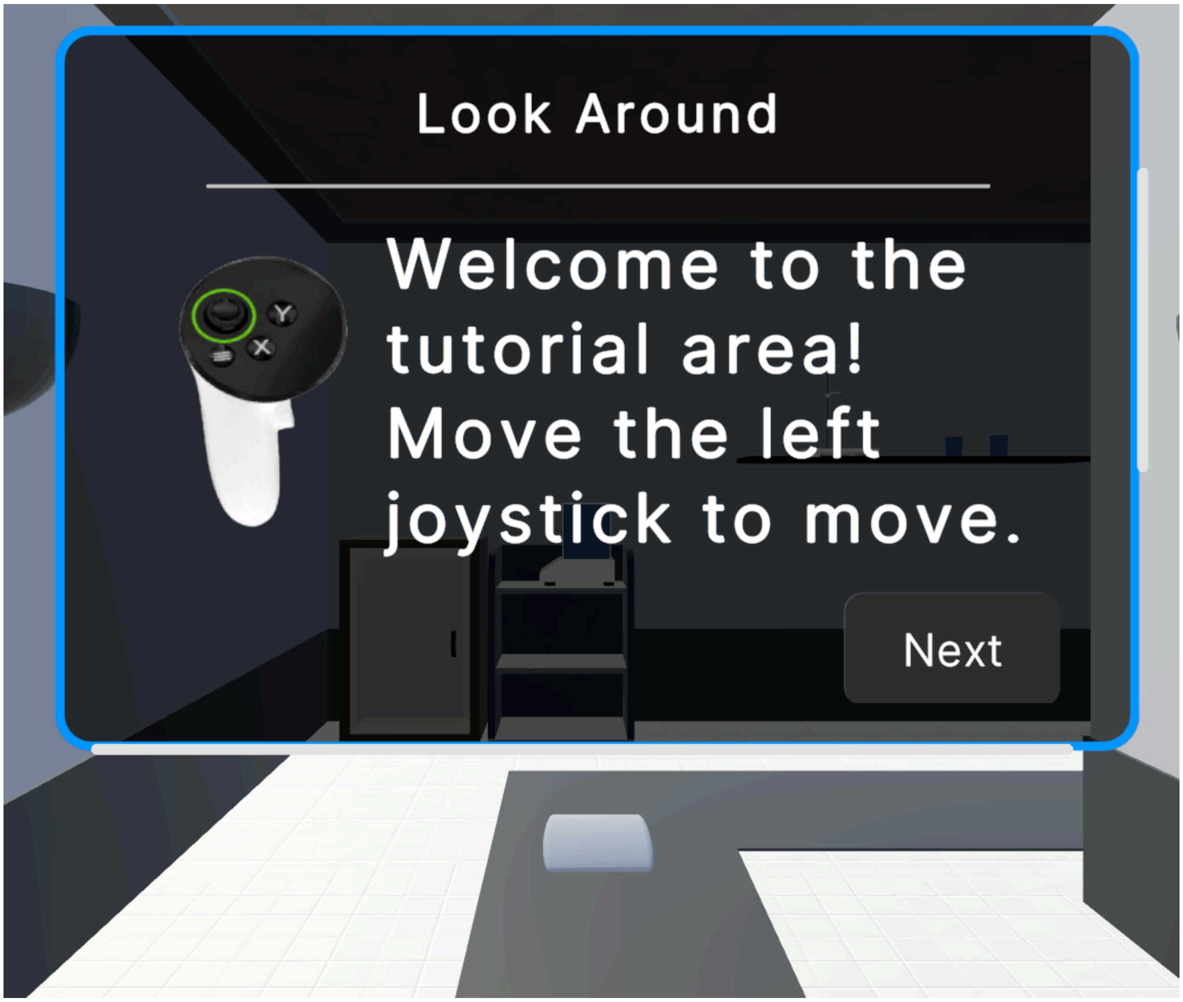
Action	Input
Movement	Left Controller Joystick
Look Around	Right Controller Joystick
Teleport	Hold the Right Controller Joystick forward, then release
Interact with UI	Left/Right Triggers
Pick up objects	Left/Right Grip Buttons
Select Bond	Hover over a bond and press the Right Trigger
UI Bank	Left Menu Button

Main Menu

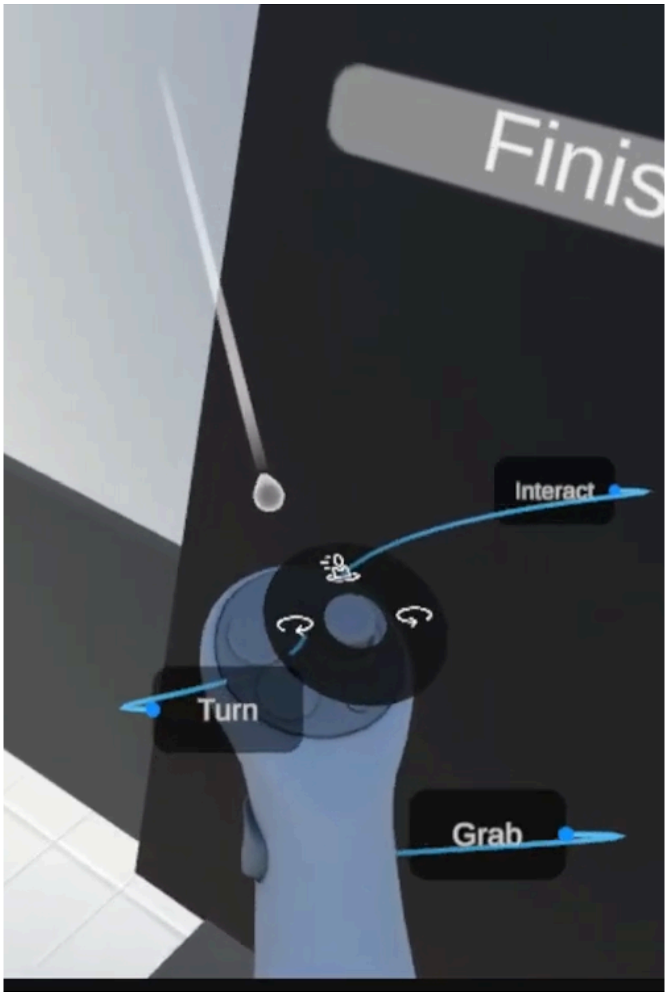
- In the main menu, players can toggle the tutorial option, adjust the music volume, or exit the app.

Tutorial Approach

- In the tutorial section, the player is walked through different UI panels to teach them general VR controls. The tutorials include walking, looking around, interacting with objects, and opening the UI bank.



- In addition to the UI panels, players can also look at the controller visuals in-game to display toolkits that indicate which buttons are which and what they do.



Main Hub

- In the main hub, players are presented with a UI panel that contains details about the app, which rooms to go to, and what's in them.

- There are two main rooms in the hub: the Builder Room and the Visual Room.
- Builder Room - initially, the player is presented with UI that tells them what to do, along with diagrams for controller inputs in case they have missed the tutorial. In here, the player may build any molecule and see its corresponding Newman Projection. Using the right trigger will specify which bond to display in the diagram.
- A UI panel to the right describes what Newman Projection is, while the diagram is shown to the left.

- Visual Room - the visual room is more of a template for a specific molecule. In this case, the player is presented with 1-bromo-2-chloropropane. This room contains details about what the molecule is, what it's composed of, and its Newman diagram.
- Unlike the Builder Room, the player can only build the specified molecule using the template. Completing the template will spawn the same molecule on the table to the left, allowing the players to view its Newman Projection.

Work Discussion

Successes

- A primary success for our project was a working molecule builder within our scope and expectations for a molecule builder
- The Newman projection aspect of the molecules and the molecule's bond selection aligned with our initial expectations
- According to the data gathered, our usability seemed to be an overall success, as the data leaned towards the higher end
- The VR comfort can be seen as a success; there may have been some discomfort in VR due to what VR itself is, however, the general consensus was positive.

Limitations

- A Primary limitation of Virtual Vials is its limited implementation of atomic properties. Full Visualization of a molecule would require a comprehensive implementation of the builder, which, was not feasible in our time frame
- A second limitation of the application was our implementation of the Newman projection. To fully utilize and visualize a Newman projection rotation around a bond for conveying different conformations, and energy states of a molecule would be need, which, also did not fully get implemented
- Lastly, our testing group consisted solely of computer science students, which limited our testing only to VR system implementation, excluding testing of the chemistry content which is a major component of our research goal.

Future Work

- The main thing to move towards an implementable system would be testing with a chemistry group to analyze if the chemistry content of the system is applicable in creating a positive contribution to a learning environment.
- Aside from user testing the main things to continue work in are the two main system upgrades: more atomic properties, and implementation of rotating a bond in the Newman projection.

Contributions to Community

Virtual Vials aims to help its users accomplish the following:

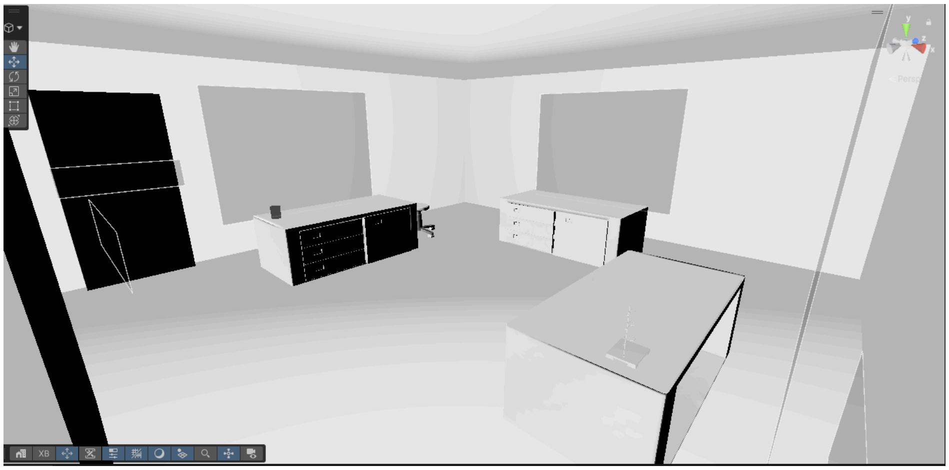
- Let users view molecules in 2D or 3D. With Virtual Vials, students will be able to view molecules from different perspectives that would be hard to do mentally.
- Virtual Vials is a VR tool which for some users, gives them that game feeling. Users may find learning in Virtual Vials more engaging than the traditional 2D format. This might help them with motivation to learn concepts more easily, and maybe even quicker for some.
- Virtual Vials fills in the gap between textbook materials and physical labs. By putting them in a lab setting, this gives them that sense of immersion, which provides them with some level of real experience.

Timeline
Timeline Summary

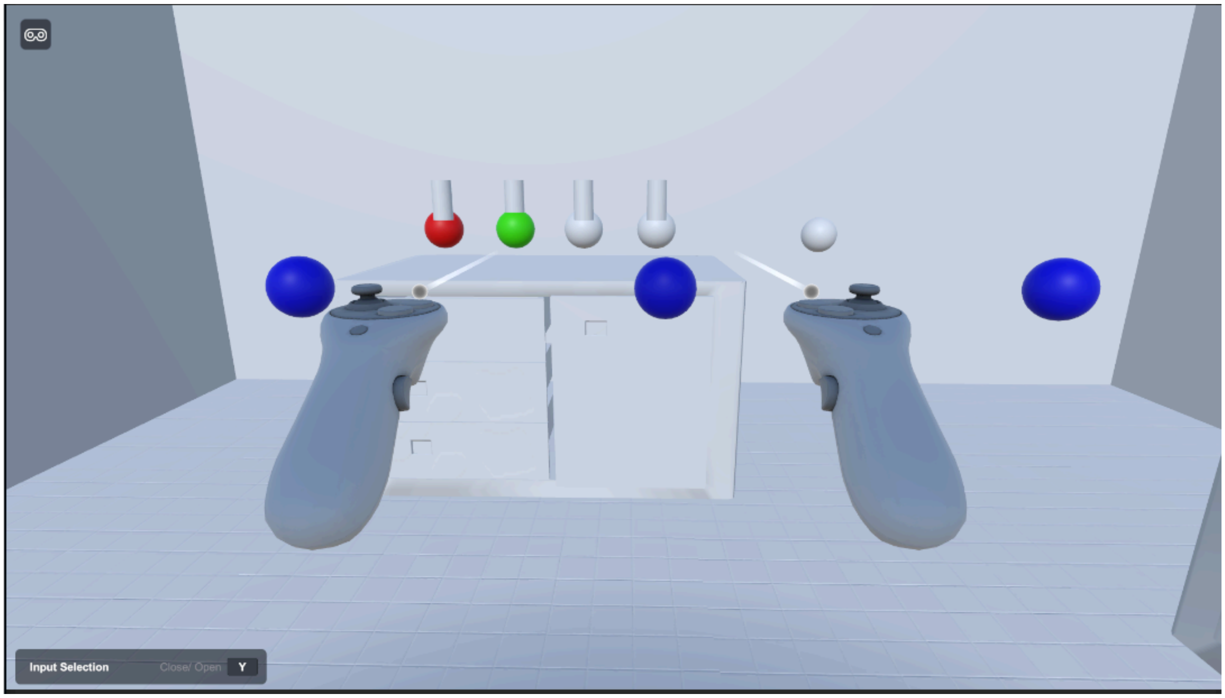
Weeks Of Development	Features to Complete
W2	- Introduction - Find research papers
W3	- Start design document - Find more research papers - Formulate testing questions - Start of Unity development
W4	- Unity VR development - Build a chemistry room - Add basic assets to interact with - Start with Start Menu - Proposal presentation
W5	- Get the most interactable/movable objects in the room - Start tutorial room
W6	- Implement UI/UX - Do sounds for interactable objects - Start with chemical interactions - Finish tutorial
W7	- Basic prototype finished
W8	- Project polishing
W9	- Project polishing
W10	- Adding last touches
W11	- Final prototype finished
W12	- Conducting tests
W13	- Statistics on test results - Finalizing project - Preparing for Presentation
W14	- Finalizing project - Preparing for Presentation

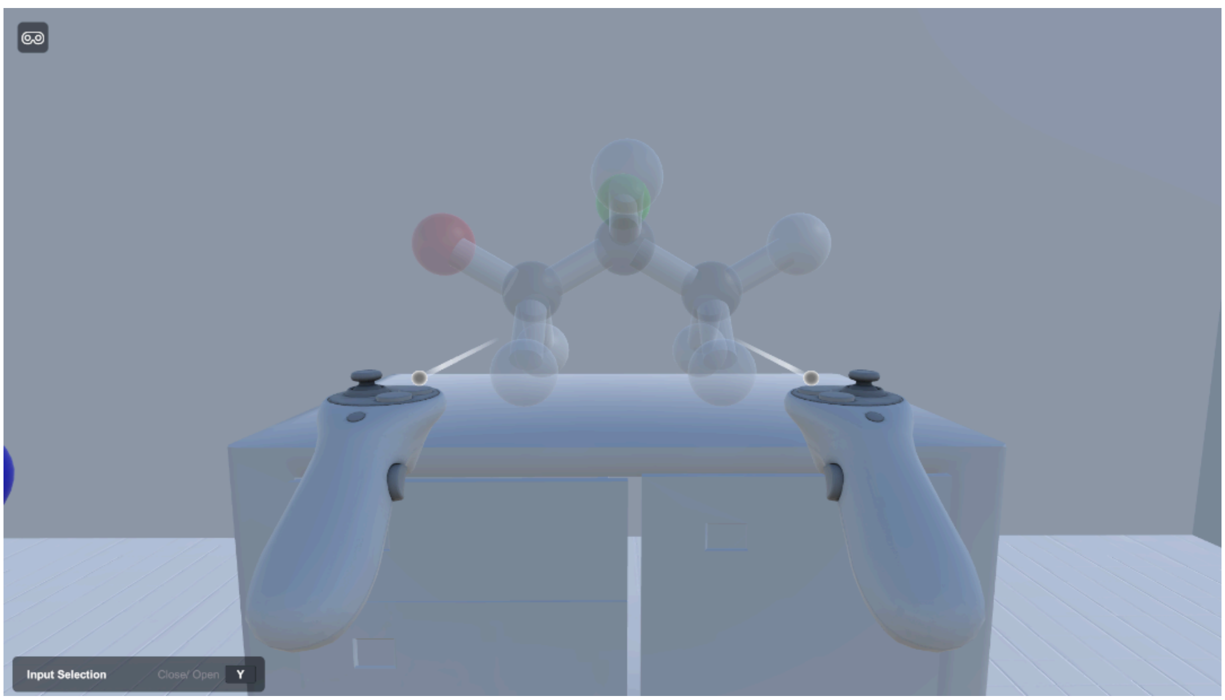
Work History

- The first few weeks were focused on planning, doing research, and learning Unity.
- Week 6: Basic implementations of the main hub room and the start of a learning room



- Week 8: Early implementation of a tutorial room to walk the users through general VR controls. Additionally, the first version of the Builder was developed.


Input Selection
Close/ Open
Y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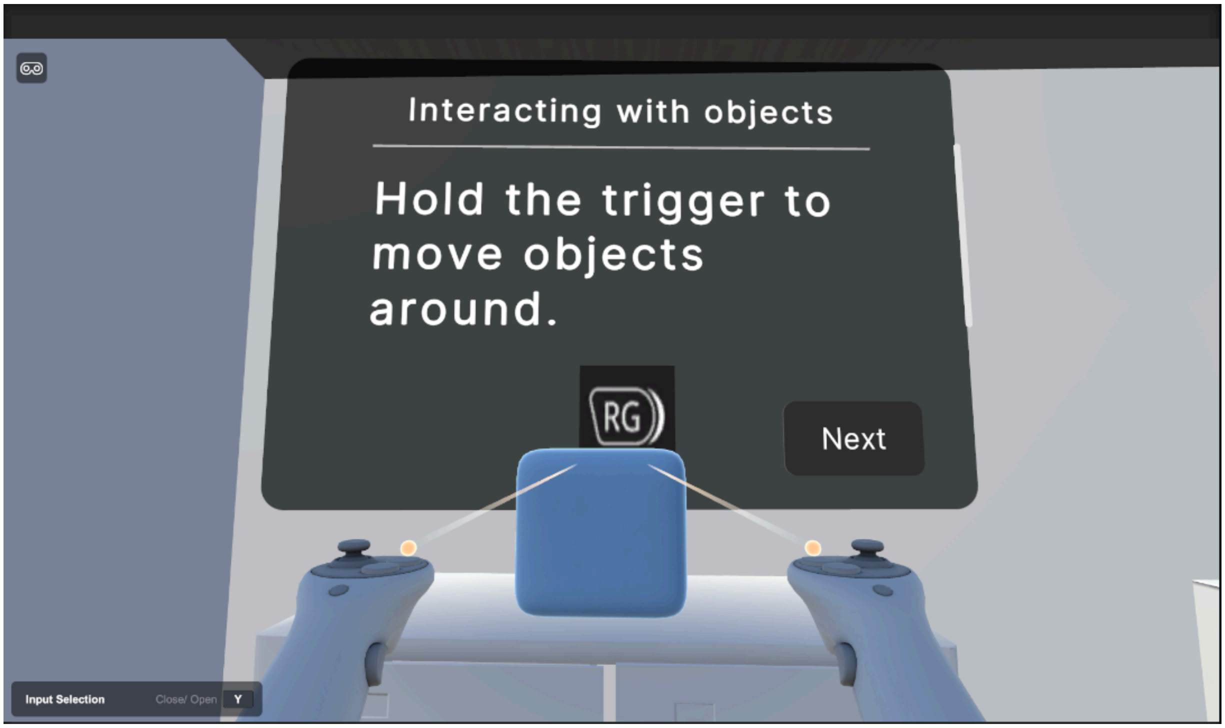
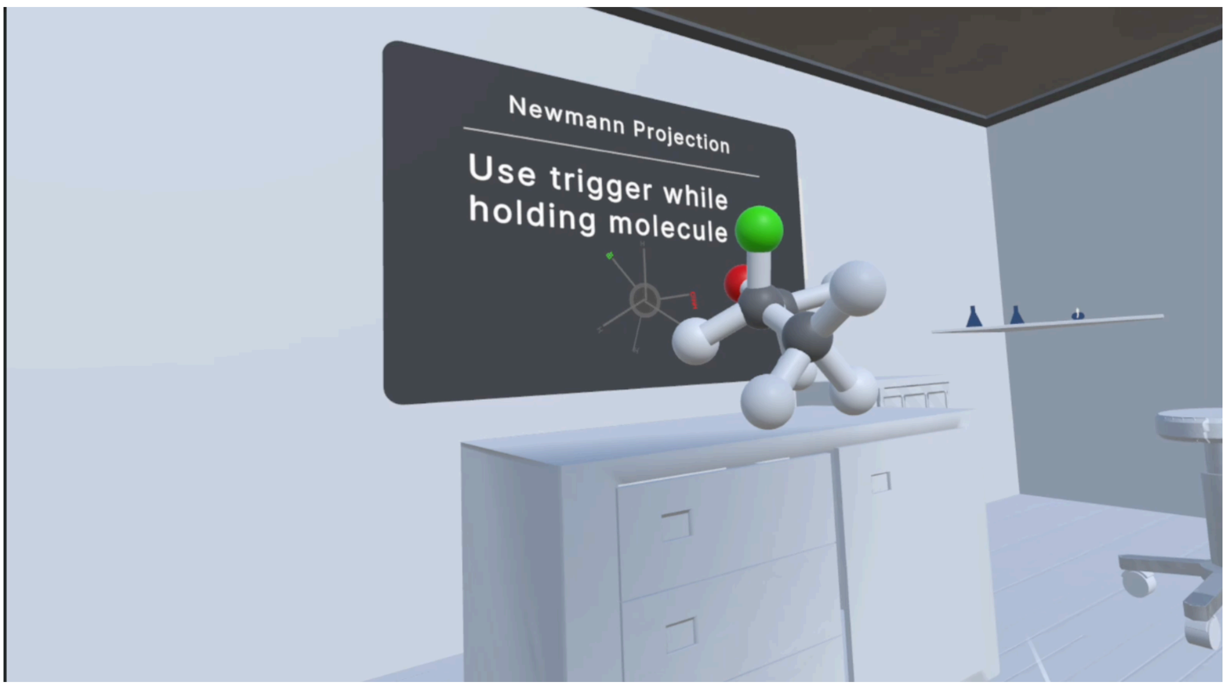
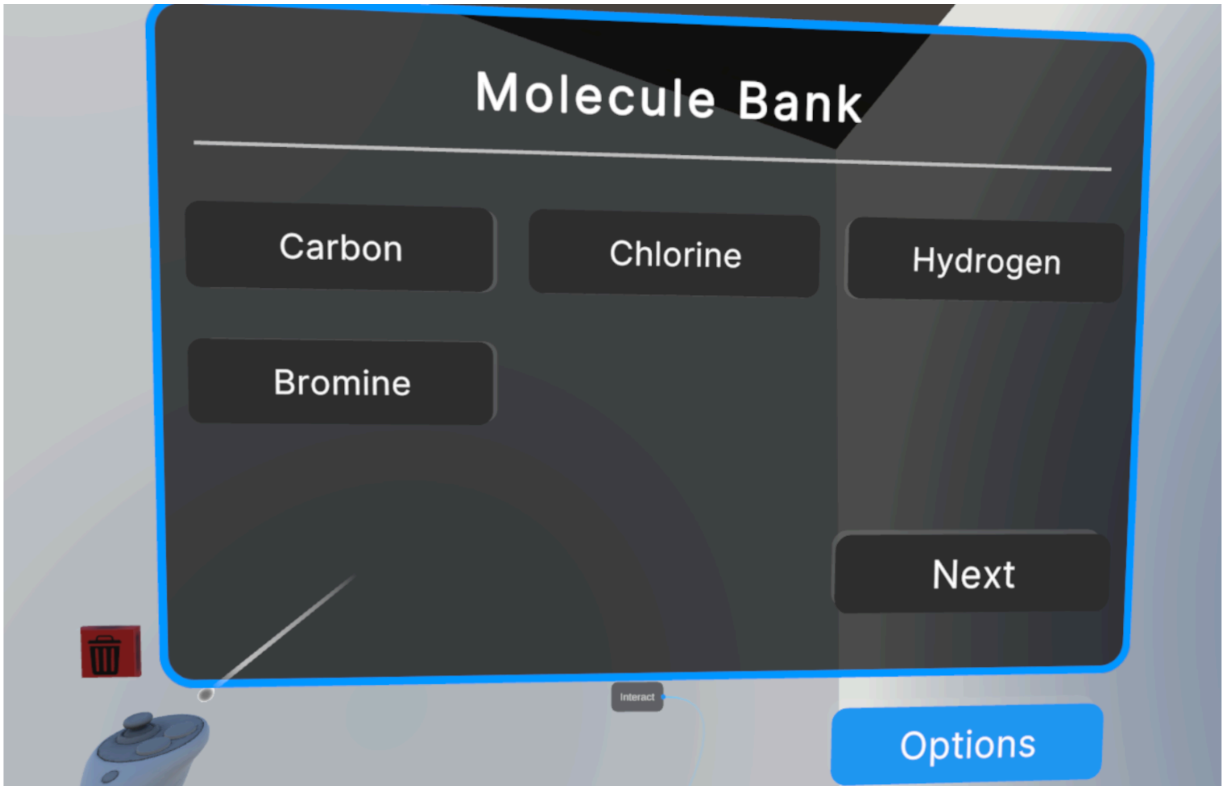
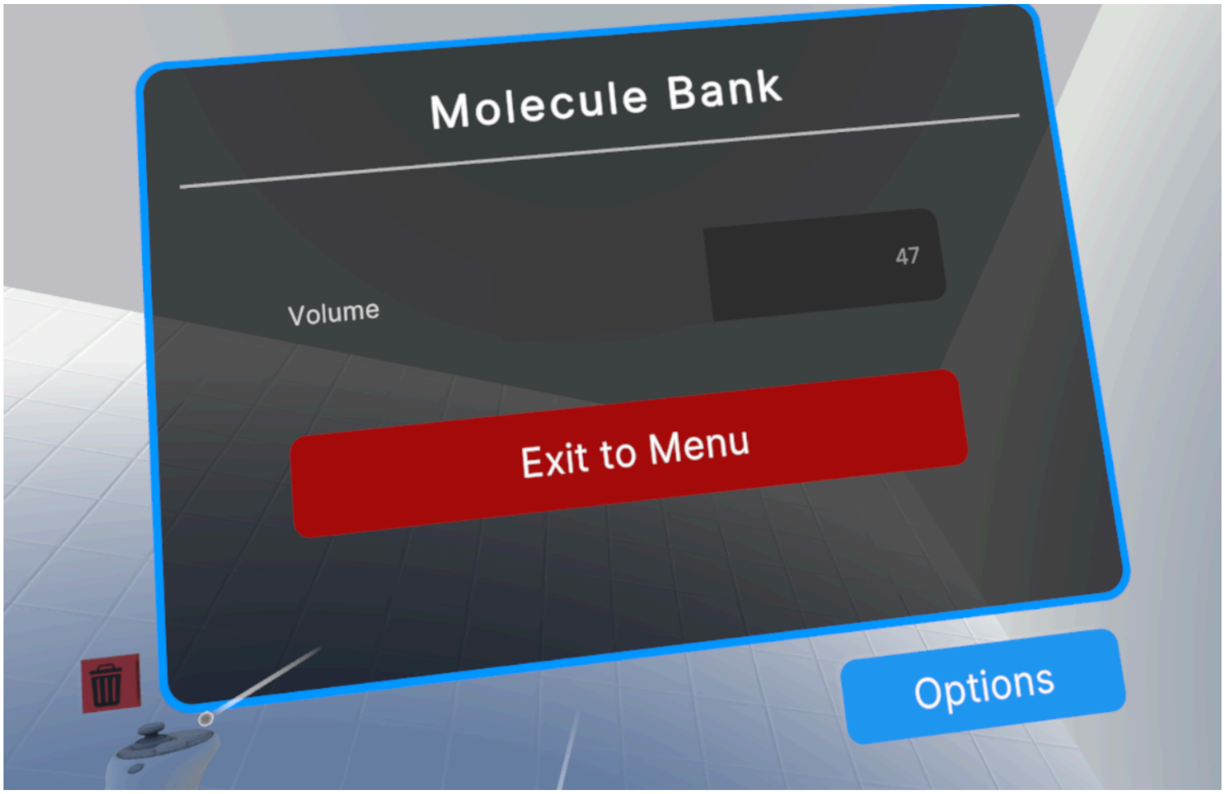


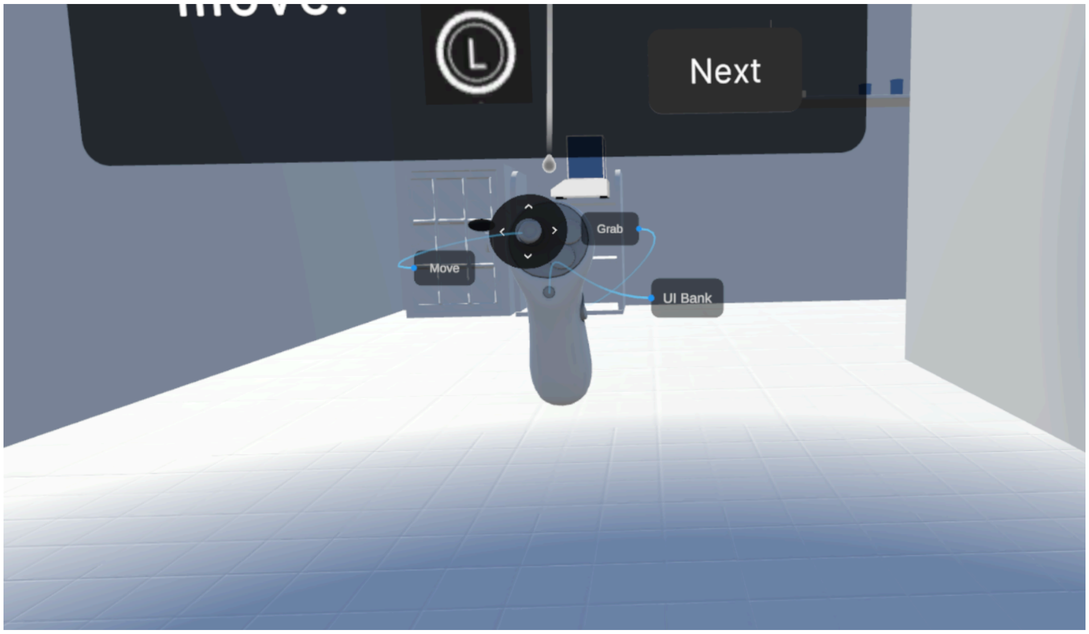
- Week 10: The main hub and rooms were fleshed out more with UI panels detailing the molecule template, 1-bromo-2-chloropropane. The first version of the Newman projection was also implemented.



- Week 11: Implemented a molecule bank with working spawns and deletions of molecules. Also added toolkits to inform users which buttons are which and what they do.


Molecule Bank
Carbon
Chlorine
Hydrogen
Bromine
Next
Interact
Options


Molecule Bank
Volume
47
Exit to Menu
Options



- Week 12: Molecule builder, general UI for volume settings and menu buttons, Molecule bank polishes
  Tutorial:
  https://youtu.be/wk0YmBJfi6o
  Builder:
  https://youtu.be/VSTbQWeal80

- Week 13: Contained General polishing with the molecule builder and a finalized first implementation of the Newman projection from the builder. Also had some finalized connections between the UI elements and the spawners for the builder.
  Newman projection work:
  https://youtu.be/ijMYIveALMk

- Week 14: Mainly consisted of user testing, but also had quite a few small bug fixes to UI, movement, and general usability of the system.